

IoT-Based Aquaponics Monitoring and Control System

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Abstract

The development of modern agriculture requires efficient and sustainable systems capable of maintaining optimal environmental conditions for plant growth and water ecosystems. Aquaponics is considered one of the most promising agricultural technologies because it integrates aquaculture and hydroponics into a single ecosystem where fish waste provides nutrients for plants while plants help filter and purify the water. However, maintaining stable environmental parameters such as water level, temperature, soil moisture, and pH is crucial for ensuring system stability and productivity. Manual monitoring of these parameters can be inefficient and may lead to delayed responses when abnormal conditions occur. Therefore, this project proposes an Internet of Things (IoT) based aquaponics monitoring and control system that automatically monitors environmental parameters and controls system components in real time. The developed system uses multiple sensors including a pH sensor, temperature sensor, soil moisture sensor, and ultrasonic sensor to measure important environmental conditions. An Arduino Pro Mini microcontroller processes sensor data and controls actuators such as a water pump, ventilation fan, RGB LED indicator, and LED display module. In addition, the ESP8266 WiFi module enables wireless communication and allows sensor data and alerts to be transmitted to users remotely via the internet. The experimental prototype demonstrates that the system can effectively monitor environmental conditions and automatically activate appropriate control mechanisms to maintain system stability. The proposed system reduces manual monitoring requirements, improves efficiency, and contributes to the development of smart agriculture using IoT technology.

1 Introduction

Agricultural production is facing many challenges such as climate change, decreasing agricultural land, and the increasing demand for food due to rapid population growth. Traditional farming methods often require large amounts of water, fertilizers, and pesticides, which can lead to environmental pollution and unsustainable resource consumption. As a result, researchers have been exploring alternative agricultural technologies that are more sustainable and efficient. According from sciencedirect the paper presents the integration of the Internet of Things (IoT) with predictive analytics to enhance the efficiency and management of aquaponics systems, which combine aquaculture (fish farming) and hydroponics (soilless plant cultivation) in a closed-loop ecosystem. In such systems, fish waste is converted into nutrients for plants, while plants help filter and recycle the water back to the fish tanks, thereby optimizing resource usage and reducing water consumption. However, due to the complexity of these systems—requiring continuous monitoring of parameters such as temperature, pH, oxygen levels, and water flow—traditional management approaches based on the automation pyramid (including SCADA, MES, and ERP) are no longer sufficient to handle large-scale, data-intensive operations. To address these limitations, the study proposes enhancing the traditional automation architecture by integrating IoT and cloud-based predictive analytics. IoT enables real-time data collection through sensors and facilitates seamless communication between different system components. The collected data is stored and processed in cloud platforms, where advanced analytics and machine learning techniques are applied to generate insights and support decision-making. Rather than replacing SCADA, MES, and ERP systems, IoT complements them by improving data interoperability, flexibility, and responsiveness. For example, IoT allows systems to predict fish and plant growth, optimize feeding strategies, manage energy and water usage, and determine optimal harvesting times, ultimately increasing productivity and profitability. The paper demonstrates these benefits through a case study of five aquaponics sites located in different geographical regions under the INAPRO project. By aggregating data from multiple sites, the system can build more robust and accurate predictive models. These models are evaluated using Monte Carlo simulations, showing superior performance compared to traditional models. Specifically, IoT-based models exhibit higher tolerance to variations, such as up to 15 percent changes in temperature coefficients and around 4 percent variation in fish feeding ratios, while still maintaining system efficiency. Additionally, the system supports remote monitoring and maintenance by applying rule-based diagnostics, enabling early detection of issues such as sensor failures, oxygen depletion, abnormal pH levels, or slow fish growth, and providing timely alerts and corrective actions. Despite its advantages, the implementation of IoT in aquaponics also presents several challenges. These include interoperability issues due to the lack of standardized communication protocols, high initial investment costs for IoT

devices and infrastructure, the need for reliable electricity and internet connectivity, a shortage of skilled personnel, and concerns regarding data security and privacy. In conclusion, the paper highlights that integrating IoT and predictive analytics into aquaponics systems is a crucial step toward smart and sustainable agriculture. While IoT significantly improves monitoring, control, and decision-making capabilities, it works most effectively when combined with existing systems like SCADA, MES, and ERP. This synergy enables more efficient operations, better resource utilization, and higher economic returns, although further research is needed to address implementation challenges and optimize long-term benefits. According from sciencedirect the paper presents the development of an Internet of Things (IoT)-based aquaponics monitoring system aimed at improving the efficiency and sustainability of modern agricultural practices. Aquaponics combines aquaculture (fish farming) and hydroponics (soilless plant cultivation) in a symbiotic ecosystem where fish waste provides nutrients for plants, and plants help purify water for reuse. However, maintaining a balanced environment in such systems is complex, as it requires continuous monitoring of critical parameters like water quality, temperature, pH, and nutrient levels. To address this challenge, the study proposes an automated IoT framework that integrates multiple sensors, including TDS, pH, and temperature sensors, connected to a microcontroller such as NodeMCU ESP8266. These sensors collect real-time environmental data, which is then transmitted to cloud-based platforms or mobile applications (e.g., Blynk) for remote monitoring and control. The system enables users to observe and manage aquaponics conditions from anywhere, reducing manual labor and improving accuracy in maintaining optimal conditions for both fish and plants. By leveraging IoT, the system can detect abnormalities in water quality and provide timely alerts, ensuring that corrective actions can be taken quickly to prevent damage to the ecosystem. This leads to healthier fish growth and improved plant yield, while also saving time and operational costs for farmers and hobbyists. Furthermore, the integration of IoT enhances data collection and analysis capabilities, allowing long-term monitoring and potential optimization of aquaponics systems based on historical trends. Overall, the research demonstrates that IoT-based automation plays a crucial role in modern smart agriculture by increasing efficiency, reliability, and scalability. The proposed system not only simplifies aquaponics management but also contributes to sustainable food production by optimizing resource use and reducing human intervention. According from researchgate the study focuses on developing an Internet of Things (IoT)-based aquaponics management system that can adapt to climate change and improve food security. With the rapid growth of population, decreasing agricultural land, and the impact of climate change and the COVID-19 pandemic, the demand for sustainable food production has increased significantly. Aquaponics, which combines fish farming and plant cultivation in a closed ecosystem, offers an effective solution where fish waste provides nutrients for plants, and plants help purify the water. However, maintaining a balanced environment

in such systems requires continuous monitoring, which can be time-consuming if done manually. To address this, the researchers integrated IoT technology to automate the monitoring and control processes. The system uses sensors to measure important water quality parameters such as pH, temperature, and dissolved oxygen. These data are processed by microcontrollers, which automatically activate actuators to regulate the system when abnormalities are detected. Users can monitor and control the system remotely through mobile and web applications, while data are stored in a cloud-based repository for analysis. The system is also powered by solar energy, making it environmentally friendly and sustainable. The development followed the Rapid Application Development (RAD) model, including stages such as analysis, design, prototyping, testing, and deployment. Both hardware components like Arduino and NodeMCU, and software tools such as Arduino IDE, C++, and Blynk were utilized. The system underwent various testing phases, including unit testing, integration testing, and system testing, to ensure reliability and functionality. Evaluation results from experts and users showed that the system performed effectively, with accurate sensor readings and efficient automation features such as scheduled feeding and water control. The mobile application successfully displayed real-time data and generated reports for users. The system achieved an overall rating of 4.32, indicating high reliability, usability, and efficiency. In conclusion, the IoT-based aquaponics system is a practical and sustainable solution that can enhance agricultural productivity, reduce labor, and support food security, making it suitable for wider implementation in farms and households

2 Methods and Materials

A. System Model and Block Diagram

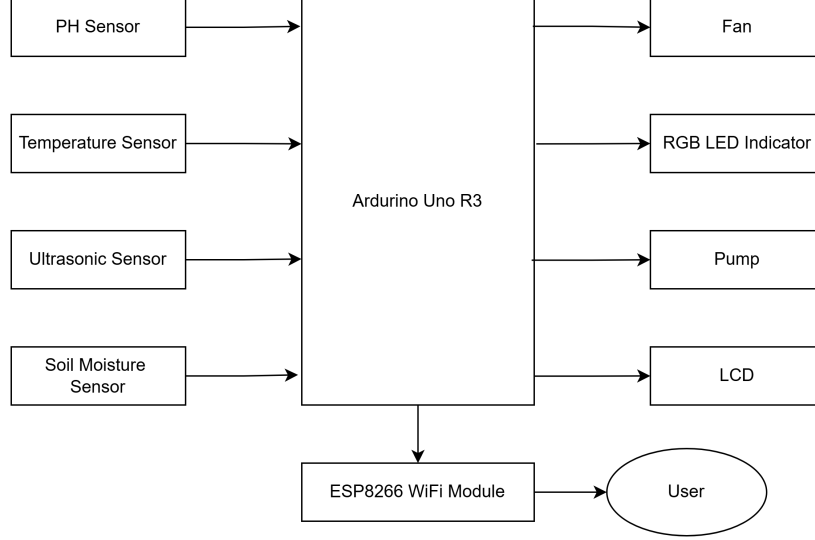


Figure 2: System Programming Flowchart

The proposed aquaponics monitoring system consists of several functional modules including environmental sensing units, a microcontroller processing unit, communication modules, and actuator control devices. Sensors such as the pH sensor, temperature sensor, soil moisture sensor, and ultrasonic sensor collect environmental data and transmit signals to the Arduino Pro Mini microcontroller. The Arduino Pro Mini processes the sensor data and determines system actions. Actuators such as pumps and fans are activated when environmental parameters exceed predefined thresholds. The ESP8266 WiFi module transmits system data to users through the internet.

B. Components and Peripheral Devices

B.1 Arduino UNO R3

The Arduino Uno R3 is a widely used microcontroller board based on the ATmega328P microcontroller, featuring multiple digital input/output pins, analog input pins, power supply pins, and communication interfaces such as UART, SPI, and I2C. In this project, the Arduino Uno R3 serves as the central processing unit of the system, responsible for collecting data from all connected sensors including the pH sensor, temperature sensor, ultrasonic water level sensor, and soil moisture sensor. The board processes the incoming data and compares it with predefined threshold values to determine appropriate control

actions. Based on the processed data, the Arduino Uno R3 controls various output devices such as the water pump, ventilation fan, RGB LED indicators, and LED display module to maintain optimal environmental conditions within the aquaponics system. Additionally, it communicates with the ESP8266 module to transmit system status and alert notifications to the user. Through real-time data acquisition, processing, and control, the Arduino Uno R3 ensures system stability, automation, and efficient operation of the overall smart aquaponics system.



Figure 3: Arduino Uno R3

B.2 ESP8266 WiFi Module

The ESP8266 module is a low-cost WiFi microcontroller widely used in IoT applications due to its built-in wireless networking capability and high processing performance. The module typically includes several important pins such as VCC, GND, TX, RX and GPIO pins. In this system, the ESP8266 is connected to the Arduino Pro Mini through serial communication using the TX and RX pins. The primary function of the ESP8266 in this project is to transmit sensor data collected by the Arduino microcontroller to remote users through the internet. When abnormal environmental conditions are detected, such as low water level or extreme temperature, the ESP8266 sends notifications to users through the IoT platform. This enables real-time remote monitoring and allows users to respond quickly to potential problems in the aquaponics system.



Figure 4: ESP8266 WiFi Module

B.3 pH Sensor

The pH sensor is used to measure the acidity or alkalinity level of the water in the aquaponics system. Maintaining a stable pH level is essential because both fish and plants require specific pH conditions to survive and grow effectively. The pH sensor typically consists of a probe electrode and a signal conditioning module. The module includes pins such as VCC, GND, and Analog Output. In this system, the analog output pin is connected to one of the analog input pins of the Arduino UNO R3. The Arduino continuously reads the pH value and displays it on the LED display module. If the measured pH value exceeds the predefined safe range, the system sends an alert to the ESP8266 module, which then notifies the user. This continuous monitoring helps maintain the chemical stability of the aquaponics environment.



Figure 5: pH Sensor

B.4 Temperature Sensor

The temperature sensor is responsible for measuring the environmental temperature within the aquaponics system. Temperature is a critical factor affecting both plant growth and fish metabolism. The sensor module includes pins such as VCC, GND, and a signal output pin that transmits temperature data to the Arduino microcontroller. When the detected temperature exceeds the predefined threshold, the system automatically activates the ventilation fan to improve air circulation and reduce heat buildup. If the temperature remains outside the safe range even after the fan is activated, the ESP8266 module sends a warning notification to the user through the internet. This automated control mechanism helps maintain optimal temperature conditions for the aquaponics system.



Figure 6: Temperature Sensor

B.5 Ultrasonic Sensor

The precision ultrasonic sensor is used to measure the water level in the aquaponics tank. The sensor operates by emitting ultrasonic sound waves and measuring the time required for the echo signal to return after reflecting from the water surface. The module typically includes four pins: VCC, GND, TRIG, and ECHO. The TRIG pin sends an ultrasonic signal, while the ECHO pin receives the reflected signal. In this system, the Arduino UNO R3 calculates the distance between the sensor and the water surface to determine the water level. If the water level falls below the predefined threshold, the system sends a warning message to the ESP8266 module so the user can take action. The water level status is also indicated using RGB LED indicators.



Figure 7: Ultrasonic Sensor

B.6 Soil Moisture Sensor

The soil moisture sensor is used to measure the moisture content of the soil where plants are grown. Maintaining adequate soil moisture is essential for plant health and nutrient absorption. The sensor typically includes VCC, GND, analog output, and digital output pins. The analog output is connected to the Arduino UNO R3 to continuously measure the soil moisture level. When the detected soil moisture drops below 40 percent, the

Arduino automatically activates the water pump to irrigate the plants. Once the soil moisture returns to the desired level, the pump is turned off. This automatic irrigation system helps ensure consistent plant growth while reducing water waste.

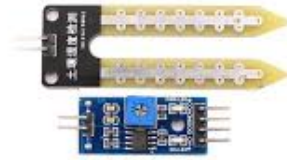


Figure 8: Soil Moisture Sensor

B.7 Relay Module

A 5V two-channel relay module is used to control high-power devices such as the water pump and cooling fan. The relay allows the Arduino Uno R3 to safely switch external devices that require higher voltage or current than the microcontroller can provide. Each channel of the relay module operates independently. The input control pins of the relay are connected to the analog pins of the Arduino. Specifically, IN1 is connected to pin A5, and IN2 is connected to pin A4. These pins are configured as digital output pins in the program. When the Arduino outputs a HIGH or LOW signal to the relay input pins, the corresponding relay channel is activated or deactivated, thereby controlling the power supply to the connected devices. In this system, one relay channel is used to control the water pump, while the other is used to control the fan. The relay module is powered by a 5V supply and shares a common ground with the Arduino to ensure proper operation.



Figure 9: Relay Module

B.8 Pump

The water pump acts as the irrigation actuator in the aquaponics system. It is responsible for supplying water to the plants when soil moisture levels become too low. The pump is controlled by the Arduino Pro Mini through a relay or transistor switching circuit. When the soil moisture sensor detects that soil moisture is below 40 percent, the microcontroller activates the pump to irrigate the plants. This automatic irrigation system ensures optimal plant hydration.



Figure 10: pump

B.9 Ventilation Fan

The ventilation fan is used to regulate temperature and maintain air circulation within the aquaponics environment. When the temperature sensor detects high temperature conditions, the Arduino Pro Mini automatically activates the fan to bring fresh air into the system and reduce heat accumulation.

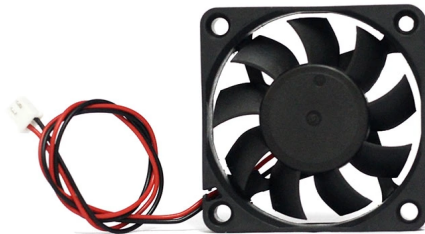


Figure 11: Ventilation Fan

B.10 RGB LED Indicator

The RGB LED module is used as a visual indicator to display system status. Different colors represent different conditions of the aquaponics system. When the water level is above the predefined threshold, the LED shows green light indicating normal operation. When the water level drops below the threshold, the LED changes to blue to alert the user. The RGB LED can also display warning conditions related to abnormal pH levels.



Figure 12: pH Sensor

B.11 LCD Display

The LCD module is used as a visual interface to display real-time pH values of the water in the aquaponics system. It receives processed data from the Arduino UNO R3 and presents the pH level clearly on the screen, allowing users to monitor water quality continuously. The LCD is connected using an I2C interface, which simplifies wiring by utilizing only four pins: GND connected to GND, VCC to 5V, SCL to pin A5, and SDA to pin A4 of the Arduino UNO R3. This configuration ensures stable and efficient data communication between the microcontroller and the display. Additionally, the LCD module includes a built-in potentiometer that allows adjustment of screen contrast for better visibility. Besides displaying pH values, the LCD can also be programmed to show warning messages when the pH level exceeds or falls below the acceptable range, thereby enhancing the system's monitoring capability and user awareness.



Figure 13: LCD Display and I2C module

C. Software Programming

The software of the system is developed on the Arduino platform and is responsible for sensor data acquisition, system control, and communication with the ESP8266 module. The `setup()` function initializes all components, including serial communication, input/output pins, relays, and the LCD display, ensuring the system starts in a safe state. The main `loop()` continuously executes key tasks such as receiving commands from ESP8266 via `handleSerialInput()`, reading sensor data through `readSensors()`, processing control logic in `updateOutputs()`, updating real-time information on the LCD using `updateLCD()`, and sending telemetry data back to the ESP8266 with `sendTelemetry()`. The `readSensors()` function collects and processes data from multiple sensors, including ultrasonic (water level), temperature, soil moisture, and pH sensors, while also determining alert conditions. Based on this data, `updateOutputs()` controls the pump, fan, and RGB LED according to either automatic thresholds or manual commands from the user. The system communicates using JSON format, where `handleSerialInput()` receives incoming data and `applyIncomingJson()` updates system parameters such as thresholds, modes, and device states accordingly. Supporting functions like `setRGB()` and `writeRelay()` handle hardware-level control of LEDs and relays. Overall, the software ensures real-time monitoring, intelligent control, and seamless integration with IoT communication. Communication between Arduino and ESP8266 is implemented using serial communication via TX and RX pins.

D. Programming Flowchart

The system is built on the ESP microcontroller platform, which functions as a centralized processing unit (CPU) that executes a continuous scanning cycle of peripheral sensors to automate crop care. The operational workflow begins by harvesting data from the

pH sensor, a vital parameter for determining the health of the soil or nutrient solution. If the pH value falls into a critical zone—either below 6 or above 8 the system immediately responds by toggling the RGB LED to green, providing an instantaneous visual alert that allows the operator to identify chemical imbalances without manually checking raw data. Simultaneously, the system employs an ultrasonic sensor to monitor the water level in the reservoir in real-time. A precise control logic is established: when the water level exceeds 10cm, the RGB LED switches to blue, serving as an overflow prevention mechanism or a "tank full" indicator. Following this, the focus shifts to soil health via the moisture sensor. A standout feature of this design is the pump control algorithm based on a hysteresis principle: the pump is only activated when soil moisture drops below 40% and remains operational until the soil reaches an ideal saturation of 70%. This mechanism not only conserves water resources but also protects the pump motor from frequent "on-off" cycling caused by signal noise. In parallel with water regulation, the system manages the microclimate through a temperature sensor. Should the environment experience extreme fluctuations—exceeding 30°C or falling below 28°C the system automatically triggers a cooling fan and adjusts the LED status to warn of thermal changes. Ultimately, all captured metrics and the operational status of the actuators are aggregated and transparently displayed on a 16x2 LCD screen. This cycle repeats indefinitely, creating a self-regulating, closed-loop ecosystem that minimizes human intervention while ensuring optimal growth conditions for the plants.

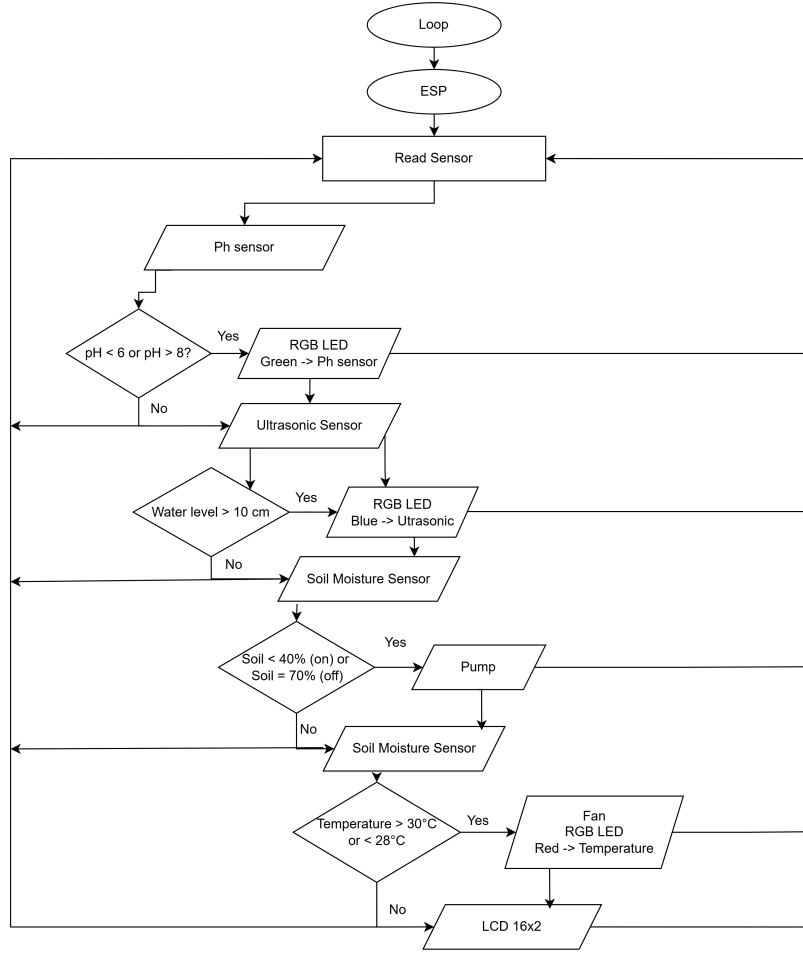


Figure 14: System Programming Flowchart

3 Conclusion

This project successfully developed an IoT-based aquaponics monitoring and control system using Arduino Pro Mini and ESP8266 modules. The system integrates multiple sensors including pH sensors, temperature sensors, soil moisture sensors, and ultrasonic water level sensors to monitor environmental conditions continuously.

The Arduino microcontroller processes sensor data and automatically activates actuators such as pumps and ventilation fans when necessary. The ESP8266 module enables wireless communication and allows users to monitor the system remotely.

The proposed system improves efficiency in aquaponics management and reduces manual monitoring efforts. Future improvements may include cloud data storage and mobile applications for enhanced monitoring capabilities.

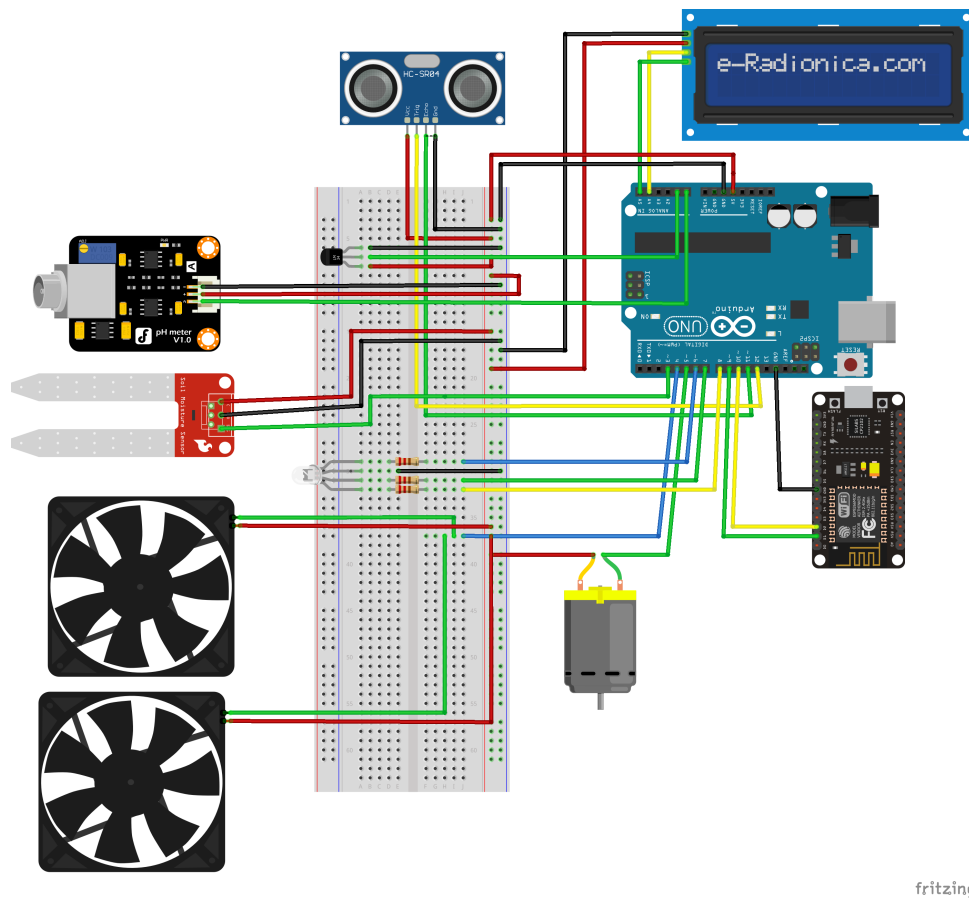


Figure 15: System Block Diagram

Table 1: BETWEEN ARDUINO UNO AND ITS COMPONENTS

Arduino UNO	Sonic Sensor	DHT11 Sensor	pH	ESP	Soil Sensor	Pump	RGB LED	Fan	LCD
GND	GND	GND	-	GND	GND		GND		GND
5V	VCC	IN	+		VCC	IN		IN	VCC
Pin 3					D0				
Pin 4								OUT	
Pin 5						OUT			
Pin 6							OUT		
Pin 7							OUT		
Pin 8							OUT		
Pin 9				D1					
Pin 10				D2					
Pin 11	ECHO								
Pin 12	TRIG								
Pin A0			A						
Pin A1		IN							
Pin A5									SCL
Pin A4									SDA

IOT102t FINAL PROJECT - GROUP MEMBER CONTRIBUTION				
Class	Class Name	Group	Group No.7	Contribution
#	Student ID	Student Name	Task	%
1	SE190861	Truong Thien Truong	Hardware, Write report	30%
2	SE190804	Tran Xuan Khoa	Software, Write report	30%
3	SE190960	Bui Le Tan Dat	Support hardware, Write report	20%
4	SE191003	Le Dinh Quy	Support hardware, Write report	20%
Total				100%

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